

Manual-Based Automated Maneuvering Decisions for Air-to-Air Combat

Kwangjin Yang,^{*} Songhyon Kim,[†] and Younggun Lee[‡]

Republic of Korea Air Force Academy, Cheongju 28187, Republic of Korea

Changyoung Jang[§]

Realtimevisual, Seoul 03925, Republic of Korea

and

Yong-Duk Kim[¶]

Agency for Defense Development, Daejeon 34186, Republic of Korea

<https://doi.org/10.2514/1.1011234>

A novel new air combat algorithm is proposed, which is based on the knowledge extracted from the experience of human pilots. First, to implement a fighter that maneuvers based on manual control, the maneuver form of the fighter is analyzed and represented as a block. Second, the blocks for each function are connected based on their relationship, and a flow diagram is presented according to the engagement situation of the adversary and ownship. Third, a behavior tree model is applied as a decision-making model to implement the flow diagram as a simulation program. The behavior tree offers good scalability because nonleaf nodes can be added when sophisticated and complex decision-making is required. The proposed method has the advantage of making all maneuvers performed by the algorithm understandable and interpretable. Additionally, it can replace expensive and dangerous dogfighting training for student pilots because the proposed model can emulate maneuvers that manned pilots would perform. To verify the proposed method, the evaluation criteria from the AlphaDogfight Trials are equally applied in the simulation. The experimental results demonstrate that the proposed method has superior engagement capability as compared to the existing air-to-air combat models.

Nomenclature

a	=	acceleration vector
k	=	gain parameter
N	=	navigation constant of proportional navigation guidance
V	=	velocity vector
$\dot{\lambda}$	=	change rate vector of line of sight
μ	=	angle vector between the velocity vector and the viewing angle vector

I. Introduction

THE development of air warfare is heading toward a future where manned pilots will lead teams of unmanned aerial vehicles (UAVs), as demonstrated by projects like The Boeing Company's Loyal Wingman in collaboration with the Royal Australian Air Force. Under the command of human pilots, UAVs will carry out various operational missions such as fighter sweep, which involves searching for and attacking enemy aircraft, and suppression of enemy air defense, which aims to neutralize or destroy surface-based air defense systems. The human pilots will focus on situational awareness and control, whereas the UAVs handle the direct execution of missions. Notably, the Defense Advanced Research Projects Agency (DARPA) is also

engaged in the development of the Air Combat Evolution (ACE) program, which explores collaborative aerial combat between manned and unmanned vehicles.^{***} As part of the ACE program, an air-to-air dogfighting competition took place in August 2020, pitting artificial intelligence (AI)-controlled UAVs against human veteran pilots. The outcome of the competition revealed the potential of AI pilot technology because the human pilots were decisively defeated.

Research into autonomous air combat has been ongoing, with early studies dating back to the 1950s when Isaacs [1] sought to model and solve air warfare via pursuit–evasion games. In recent years, extensive research has focused on applying AI technology to UAVs. However, the application of AI in this context requires rigorous verification processes before it can be integrated into actual systems, primarily due to the complex and high-speed nature of fighter jets operating in three-dimensional (3-D) space. Even minor errors in such systems could have catastrophic consequences, making thorough verification essential for practical applications beyond other domains.

In this paper, we propose a method for unmanned combat technology to perform autonomous air combat based on the Basic Employment Manual (BEM) [2] knowledge, which serves as a reference book for fighter pilot training. The BEM is a document that describes the tactics, techniques, and operating procedures of air force fighters, providing the fundamental maneuvering procedures and skills required to complete various missions. It includes basic fighter maneuvers (BFMs), which deal with maneuvering locations and how to fly in space, providing a solid foundation and knowledge to develop effective tactics for making the right decisions in tactical missions. Although many studies have been conducted on autonomous aerial combat in within-visual-range (WVR) air combat situations, our proposed framework is the first to extract knowledge from tactical manuals and utilize them as basic data for fighter maneuvers.

The main contribution of this study is the composition of the actual pilot's knowledge and decision-making process in the form of a flow diagram. The conventional knowledge-based air combat algorithm research using BFMs focuses on combinations of basic movements

Received 2 January 2023; revision received 9 July 2023; accepted for publication 30 August 2023; published online Open Access 29 September 2023. Copyright © 2023 by the American Institute of Aeronautics and Astronautics, Inc. All rights reserved. All requests for copying and permission to reprint should be submitted to CCC at www.copyright.com; employ the eISSN 2327-3097 to initiate your request. See also AIAA Rights and Permissions www.aiaa.org/randp.

^{*}Professor, Department of Mechanical Engineering, 635 Danjae-ro Namil-myeon Sangdang-gu.

[†]Professor, Department of Computer Science, 635 Danjae-ro Namil-myeon Sangdang-gu.

[‡]Professor, Department of Electronics and Communication Engineering, 635 Danjae-ro Namil-myeon Sangdang-gu; yglee2019@gmail.com (Corresponding Author).

[§]Assistant Engineer, Research and Development, 1605 Sangam-dong Mappo-gu.

[¶]Senior Engineer, Defense AI Technology Center, 160 Bukyuseongdae-ro Yuseong-gu.

^{***}"Air Combat Evolution," Defense Advanced Research Projects Agency, <https://www.darpa.mil/program/air-combat-evolution> [retrieved 24 September 2023].

or controls rather than precise maneuvers. The BFM-based method has limitations because it is challenging to benefit from maneuvering perspectives for immediate gain. In contrast, the proposed method covers maneuvers in the offensive basic fighter maneuver (OBFM), the defense basic fighter maneuver (DBFM), and the high aspect basic fighter maneuver (HABFM) encountered WVR, guaranteeing higher rewards. The proposed method is implemented as a simulation program using the behavior tree (BT). Through combat experiments with conventional models, the proposed method demonstrates its superiority. Furthermore, the proposed method allows real-time simulation of air-to-air combat, and the entire maneuver can be interpreted by humans. As a result, it can be used for student pilot training and can be easily applied to current technologies being developed by human pilots for validation.

The rest of this paper is organized as follows. Section II reviews some related research work. The BEM-based flow diagrams are proposed in Sec. III, whereas simulation implementation is explained in Sec. IV. Experimental results are shown in Sec. V, followed by the conclusions in Sec. VI.

II. Related Work

Research on autonomous air combat can be divided into three approaches: mathematics-based, knowledge-based, and learning-based approaches [3]. The mathematics-based approach has involved various studies, modeling dogfights as pursuit–evasion games [1] and attempting to solve the problem using differential game theory [4–8]. Additionally, several studies have explored solving the problem using optimization theory [9–12]. However, this approach has limitations when applied to real systems, such as simplifying the problem to a two-dimensional plane or fixed roles for tracers and avoiders. On the other hand, methods utilizing the Markov decision process, which calculates control inputs after discretizing the state space [13,14], influence diagrams applying receding horizon control with the k -step look-ahead method [15,16], and approximate dynamic programming that mitigates computation limitations [17,18] have been explored. Nevertheless, these methods still have restrictions due to the substantial computational requirements and the challenge of solving air combat in diverse situations with a limited set of discrete actions.

Early studies on knowledge-based approaches included adaptive maneuvering logic (AML), the tactical guidance research and evaluation system (TGRES), and the trial maneuver. AML [19,20] was developed based on expert knowledge and used to select actions from a set of elementary maneuvers based on the relative geometry of the ally and the adversary. Later, AML evolved into the TGRES [21,22], which improved performance by incorporating high-fidelity airplane dynamics. This method is useful for initial pilot training, but its performance model differs from reality; and the control scheme must be carefully adjusted for each aircraft type, making it time-consuming to update or modify the rules. In contrast, a study on the chess-type tree [23] constructed a look-ahead tree to conduct dogfights, allowing only one player to act at a time, similar to a game of chess, using seven elemental maneuvers. However, this method has limitations in describing numerous situations. Therefore, research is being conducted on a game matrix [24] that simulates dogfights by representing the movements of both enemies and allies in a 7×7 square matrix. However, this approach also has limitations in developing various dogfight situations with only seven actions.

In recent years, research on the utilization of learning techniques in combat simulations has gained significant popularity. One study [25] focused on creating new combat tactics using a learning classifier system, whereas another [26] explored autonomous combat in a high-performance simulation environment using a genetic fuzzy tree. More recently, researchers have actively applied reinforcement learning (RL) techniques [27–29]. In fact, a recent study [30] employing hierarchical RL secured the second place in the AlphaDogFight aerial combat contest hosted by DARPA (known as the AlphaDogFight Trials). Although the results of RL-based simulations have shown promise, implementing these systems in practice will require extensive verification processes. Moreover, interpreting the maneuvering

principles and situational judgment learned by RL-based systems may pose challenges for humans.

III. BEM-Based Flow Diagrams

A. Basic Employment Manual

This section describes the BEM [2], which serves as the primary basis for this paper. The BEM is a comprehensive reference document for air force fighter tactics, technology, and operating procedures. Its main objective is to provide the fundamental maneuvering procedures and technologies necessary to successfully execute various missions using fighters. The technical scope of the BEM encompasses all aspects, from flight preparation to takeoff, landing, and basic flight techniques for air-to-air and air-to-ground combat, as well as flight planning considerations. By offering a solid foundation and knowledge to develop effective tactics, the BEM assists pilots in making informed decisions in any tactical mission scenario. Additionally, the BEM includes BFMs, which encompass the strategies for both offensive and defensive maneuvers, guiding pilots on how to position themselves in desired locations and spaces during combat engagements.

B. Flow Diagrams of Dogfight

1. How to Make a Flow Diagram

In order to implement a fighter model that maneuvers based on the BEM, the maneuvering form of the fighter is prepared with a flow diagram and a statement for each block. Specifically, each block is labeled in consideration of the concept of the individual maneuver, and the blocks are connected in consideration of the relationship with the next maneuver decision factor. The block in the flow diagram is set to the *maneuvers* level corresponding to the third level in Fig. 1 [31].

Figure 1 is a conceptual hierarchy for air mission design and strategic decision-making, which consists of five layers. The first level is the *control* layer, the second level is the *movements* layer, the third level is the *maneuvers* layer, the fourth level is the *tactical decision* layer, and the fifth level is the *strategic decisions* layer. The statement explains the definition and maneuver concept for each block, defines the relationship and conditions between connected blocks, and specifies the detailed procedure at the control level for performing the maneuver.

Here, we explain how to create a flow diagram in the presence of figures as in Fig. 2. Figure 2a explains that the entry window changes based on 40 deg of the aspect angle (AA), and Fig. 2b explains how to check the entry cue from the pilot's point of view, as described in Fig. 2a. According to Ref. [2], one must get to the turn circle and then solve the angular and range problems presented in order to prosecute an adversary. The first critical step is to recognize the ownship's position relative to the adversary's turn circle. As depicted in Fig. 2a, when the attacker initiates the confrontation with the adversary positioned at an aspect angle of less than 40 degrees and centered on the HUD (Head-Up Display), moving directly ahead is a highly effective strategy. This approach aids in establishing the visual cues needed to handle more dynamic starting conditions in subsequent encounters. If the angle of approach between our ownship and the adversary exceeds

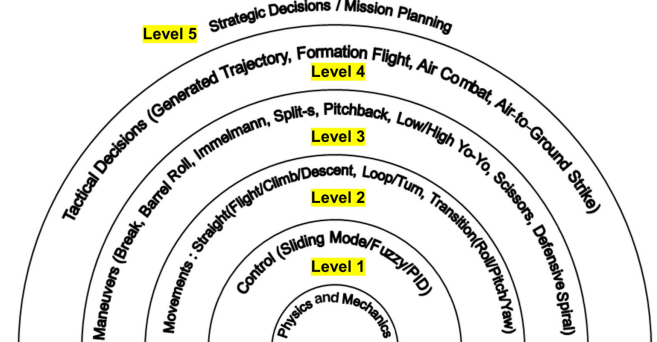


Fig. 1 Abstraction stack of air mission planning [31].



Downloaded by KYUNGPOOK NATIONAL UNIVERSITY on March 25, 2026 | http://arc.aiaa.org | DOI: 10.2514/1.1011234

Downloaded by KYUNGPOOK NATIONAL UNIVERSITY on March 25, 2026 | http://arc.aiaa.org | DOI: 10.2514/1.1011234

Downloaded by KYUNGPOOK NATIONAL UNIVERSITY on March 25, 2026 | http://arc.aiaa.org | DOI: 10.2514/1.1011234

Downloaded by KYUNGPOOK NATIONAL UNIVERSITY on March 25, 2026 | http://arc.aiaa.org | DOI: 10.2514/1.1011234

Downloaded by KYUNGPOOK NATIONAL UNIVERSITY on March 25, 2026 | http://arc.aiaa.org | DOI: 10.2514/1.1011234

Downloaded by KYUNGPOOK NATIONAL UNIVERSITY on March 25, 2026 | http://arc.aiaa.org | DOI: 10.2514/1.1011234



Downloaded by KYUNGPOOK NATIONAL UNIVERSITY on March 25, 2026 | http://arc.aiaa.org | DOI: 10.2514/1.1011234

Each block is numbered to distinguish it, and they are connected according to their relationships. A rectangular block indicates a single maneuver; a diamond-shaped block denotes a conditional block; and the next maneuver varies, depending on whether it is satisfied. Colored blocks indicate maneuvers in which sub-blocks are separately specified. All the details of the colored blocks are provided as Supplemental Material.

When viewing Fig. 3 from the top, it is divided into three main types. The HABFM is a situation in which the adversary and the ownship are neutral, the OBFM is a situation in which the ownship is dominant, and the DBFM is a situation in which the adversary has an advantage. At this time, the degree of dominance is classified according to the initial engagement setup. Specifically, the initial setup condition is decided in the “combat geometry” block (block 1) in Fig. 3; then, the situation is recognized according to the setup condition in block 2 (“situation awareness”); finally, the direction of execution is determined among the OBFM, the HABFM, and the DBFM through relative superiority in block 3 (“Superior?”). The OBFM, DBFM, and HABFM are judged by the computer at every moment in an engagement situation. During a dogfight, it is essential to consider both the geometric positional relationship between aircraft and their relative energy, including altitude and speed. The 3–9 line (as shown in Figs. 4a and 4b), which runs from the 3 o’clock position to the 9 o’clock position, is typically used to determine the geometrical positional relationship between the aircraft. A crucial factor in combat is the relative energy of each aircraft, with greater energy providing a significant advantage.

The objective of the OBFM is to shoot down the adversary as quickly as possible without wasting time and fuel. To achieve this, the pilot should maintain the control zone after entering the turn circle; then, they should use the weapon as soon as the opponent

enters the WEZ. The corresponding blocks are from block 8 to block 12. Here, the control zone is shown in Fig. 5 in the manual. To be in the control zone, three conditions must be satisfied: relative distance, relative angle, and relative speed. Only when the preceding three conditions are satisfied can the pilot maintain their superiority and win.

The HABFM is a maneuver used in a situation where neither the adversary nor the ownship has an advantage due to a high angle encounter. In blocks 14 to 19, maneuvers are repeated until one side gains an advantage. At this point, either a one-circle or two-circle maneuver is selected based on a comparison of the distance to the opponent and the energy level. In a one-circle maneuver, the adversary and the ownship turn in opposite directions, creating a single circle as shown in Fig. 4a. Conversely, in a two-circle maneuver, the adversary and the ownship turn in the same direction, with each maneuvering in a circle as shown in Fig. 4b. The one-circle and two-circle blocks are connected to block 20. Figures 4c and 4d show the details of the winning cue block used to determine who has the upper hand during one-circle or two-circle maneuvers. Specifically, block 20-1 in Fig. 4c decides who has the advantage after a 90 deg turn in a one-circle maneuver, as shown in Fig. 4a. If the red aircraft is ahead of the blue aircraft and the energy on the EM (Energy-Maneuverability) chart is similar or higher, the blue aircraft obtains the winning cue (block 20-4). If the energy is lower, it is determined to be a neutral situation (block 20-5). Upon receiving a winning cue, a two-circle maneuver is performed (block 15 in Fig. 3), and the winning cue is determined after a 180 deg turn, as shown in block 17-1 in Fig. 4d. If the AA is less than 90 deg, as in the case of the blue aircraft in Fig. 4b, it is a favorable situation; and the maneuver is switched to the OBFM (blocks 4 through 35).

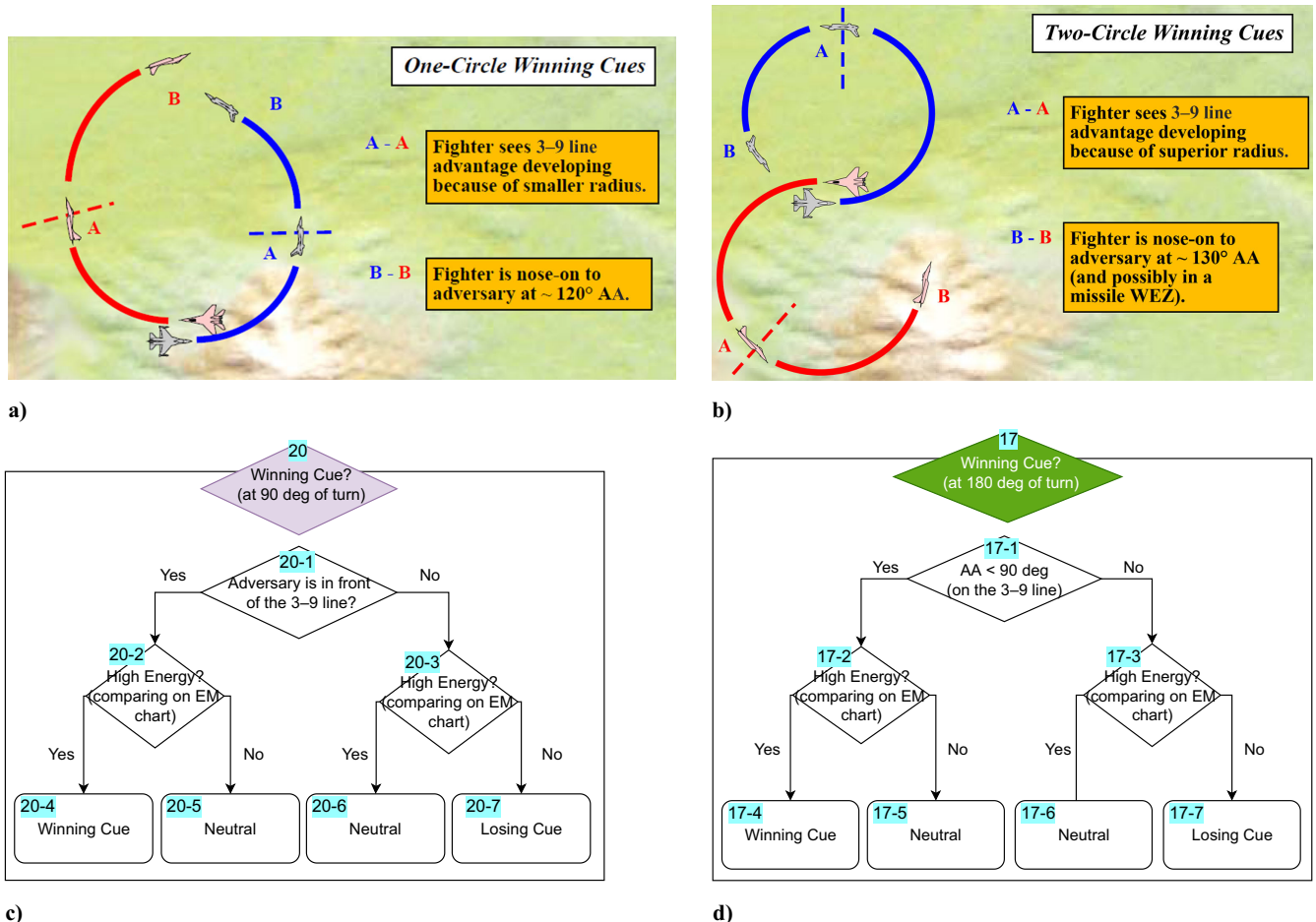


Fig. 4 Figures and matched diagrams: a) one-circle winning cue illustration, b) two-circle winning cue illustration, c) detailed blocks of “Winning cue?” of one-circle maneuver (block 20), and d) detailed blocks of Winning cue? of two-circle maneuver (block 17).

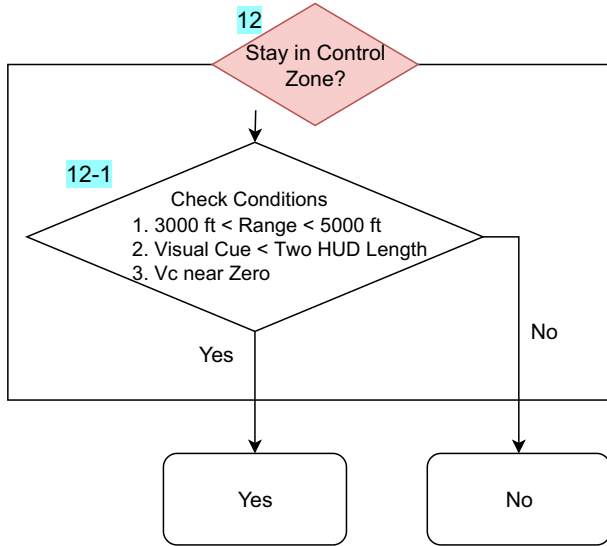


Fig. 5 Detailed blocks of “Stay in control zone?”.

Finally, the DBFM is a situation where the ownship is at an adverse situation, with the goal of surviving as long as possible from an enemy threat and turning it into a neutral or superior situation by taking advantage of the opportunity. To this end, the ownship must recognize and exploit the adversary’s mistakes and weaknesses to create geometric problems for the adversary as much as possible. Specific maneuvers are blocks 23 to 33. Block 21, Scissors, is a maneuver performed when both fighters are in the HABFM and no one has a winning cue.

IV. Simulation Implementation

A. System Structure

This section describes the structure of the system to implement and simulate the flow diagram derived in Sec. II. The system structure is largely composed of a decision-making model, an induction control model, and a flight dynamics model. The overall structure is shown in Fig. 6.

The decision-making model corresponds to the AI that automatically decides the maneuver based on the flow diagram. An easy way to implement AI is the finite state machine (FSM), which has a finite number of states. However, the FSM has a shortcoming, in that it becomes difficult to manage as the number of states increases or state transitions become more complex. A new BT has emerged to compensate for these shortcomings. Unlike the FSM, the BT defines a decision-making model in a treelike manner. The BT makes decisions at nonleaf nodes, and it executes the decisions when it reaches the leaf nodes. The BT is highly scalable, and so you can add nonleaf nodes when sophisticated and complex decisions are needed. Therefore, the proposed method implements a decision-making model based on the BT.

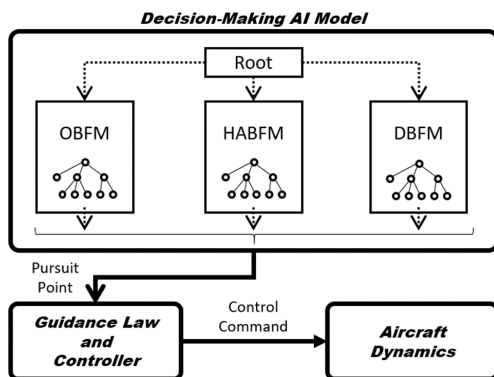


Fig. 6 Simulation system structure.

In order to carry out the maneuvering action from decision-making, a guidance and control model is needed. In this paper, a guided control model based on pursuit points is applied [32,33]. In addition, JSBSim [34], which is a six-degree-of-freedom flight dynamics model, is used for the flight dynamics model.

B. Decision Model Using Behavior Tree

The BT is largely composed of five types of nodes. First, the root node is the node representing the beginning of the tree. Every tree has a root node, and each loop traverses the tree from the root node down to the leaf nodes. Second, the composite node defines the basic rules in which the branch is executed. Composite nodes include a sequence node and a selector node in detail. A sequence node returns a value of “true” when multiple subnodes exist and the execution result of all subnodes is true. The selector node returns true to the parent node at the moment when any of the child nodes is true. Third, the task node performs an actual action. The task node is a leaf node of the tree, and it defines and uses the actions that the object actually performs. Fourth, the decorator node determines the situation and performs conditional branching. The decorator node checks the condition and, if true, executes the child node. The decorator node is attached to a composite node or task node to define which branch of the node will run on. Finally, the service node plays a role of extracting and storing information necessary for situation determination. The service is executed at a set frequency while branching to the composite node is executed. The BT can model complex and sophisticated AI by combining the characteristics of each node.

WVR during air combat, the situation changes greatly at every moment and requires an instantaneous judgment. Because it is repeated every time a tick event occurs up to a node, it is possible to model AI that has a great advantage in instantaneous situation determination. The AI decision-making model consists of tasks in the leaf node of the BT that designate the pursuit point, which is the direction the aircraft should proceed; and the upper node determines which pursuit point to track. It is designed as a decision-making part.

Figure 7a shows the overall implementation of the BT, and Fig. 7b shows the implementation of the OBFM among the BTs. Also, Fig. 7c shows an enlarged view of four nodes, which are the top nodes of the OBFM. In Fig. 7c, the first node is a decorator node, named “OBFM?” (blue background box in the figure), and the second node is a selector node. At the decorator node, if the conditions of the OBFM are satisfied, the following two services named “TurnCheck” and “SpeedControl” (green background box in the figure) are going to be executed. In Figs. 7a–7c, the line connecting the nodes is marked in thick white, which means that the node is currently active.

C. Guidance Law and Controller

When a task node is reached in the action tree, each task node must create a control of the throttle, aileron, elevator, and rudder. A pursuit point command to perform a maneuvering action is generated from decision-making. In order to direct the corresponding tracking point, the guidance algorithm is composed of the sum of the pursuit guidance technique [32] and the proportional navigation guidance [35] for agile behavior of the target, as shown in Eq. (1):

$$\mathbf{a}_{\text{cmd}}^N = k_{\text{PG}}(\boldsymbol{\mu} \times \mathbf{V}^N) + N(\dot{\lambda} \times \mathbf{V}^N) \quad (1)$$

where $\mathbf{a}_{\text{cmd}}^N$ is the acceleration command of the guidance law on the reference coordinate system, $\boldsymbol{\mu}$ is the angle vector between the velocity vector $\mathbf{V}^N$ of the vehicle on the reference coordinate system and the viewing angle vector at which the vehicle sees the pursuit point, and $\dot{\lambda}$ is the change rate vector of the line-of-sight angle, as shown in Fig. 8. That is, both the flight-path angle difference and the rate of change of the line of sight are reflected in the generation of the acceleration command. In this paper, the guidance gain k_{PG} is set to 2.0 and the navigation constant of proportional navigation guidance N is set to 0.5 through simulation. From the guidance command, the roll angle command and the vehicle’s G acceleration

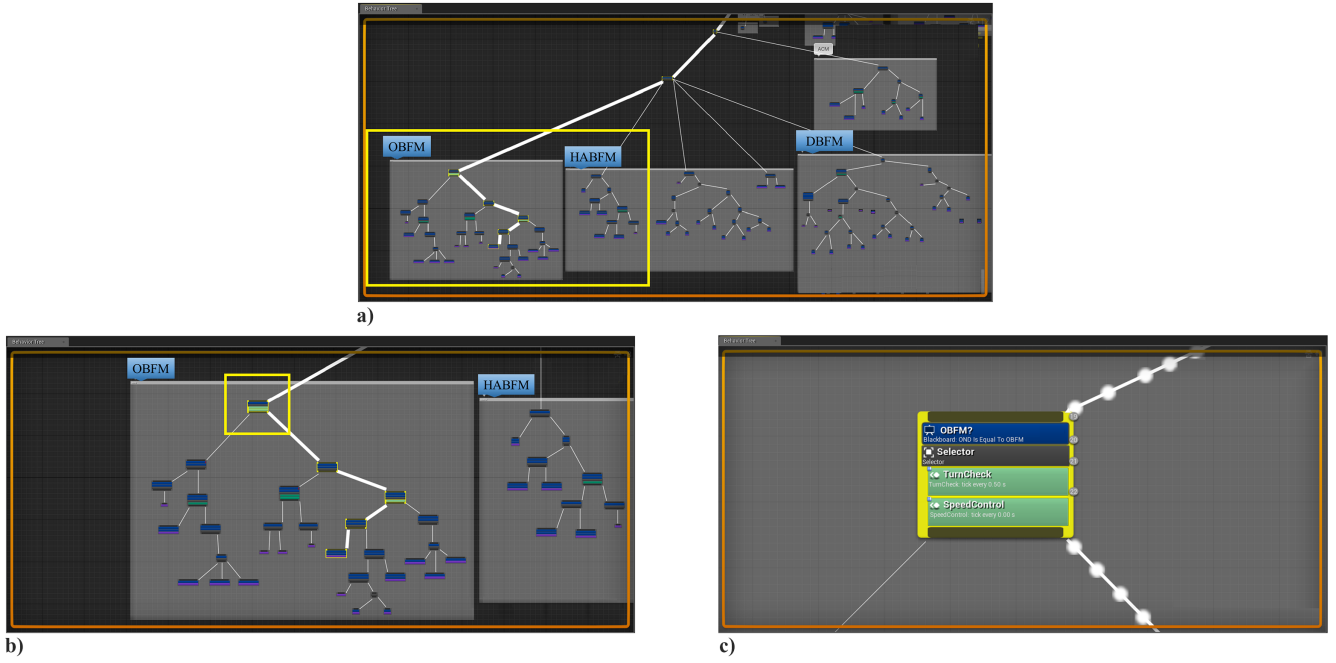


Fig. 7 Implemented behavior tree: a) entire BT (from bottom left, the node groups in charge of the OBFM, HABFM, and DBFM are shown, b) enlarged view of the yellow box shown in Fig. 7a, showing the OBFM BT, and c) enlarged view of the yellow box shown in Fig. 7b, showing the top node of the OBFM.

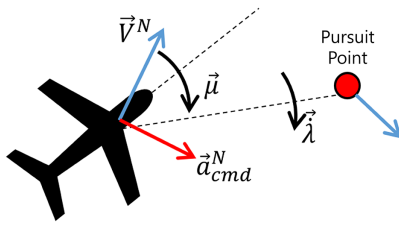


Fig. 8 Guidance law on the tacking situation.

command can be converted from the nonacceleration command considering gravity. The control law follows the roll angle command, the G acceleration command, and the speed command.

The aircraft used are F-16s, and JSBSim [34] is used as the flight dynamics model. JSBSim is an open-source flight dynamics model that is used widely in the unmanned aerial vehicle research area [36].

D. Simulation Environment

The simulation environment is built using the Unreal Engine software. Unreal Engine provides an editor for modifying the BT and a data repository called Blackboard, making BT implementation convenient.

Additionally, a three-dimensional visualization tool allows real-time monitoring of the simulation's execution status. As an example, Fig. 9a displays a top view that monitors the air battle situation between two aircraft. In this view, the red aircraft is in a lag pursuit, engaging the blue aircraft in an OBFM situation. Furthermore, Fig. 9b shows real-time monitoring from an external view, depicting the red aircraft chasing the blue aircraft. The white font text embedded in Fig. 9 indicates the heading, altitude, and speed of each aircraft; whereas the red and blue fonts represent the aircraft's identification and relative superiority, such as the OBFM or DBFM. In the bottom-right corners of Figs. 9a and 9b, the controls for the throttle, aileron, elevator, rudder, and residual fuel of the selected aircraft are displayed. These visualization tools enable the evaluation and optimization of the decision-making model by experienced pilots, ensuring its completeness.

V. Experimental Results

A. Evaluation Criteria

The effectiveness of the proposed algorithm is verified through 1:1 air-to-air combat engagements. To compare implementation methods, the extent of damage inflicted through weapon usage is calculated. Damage is considered effective when the adversary's center of gravity falls within the weapon engagement zone. As shown in Fig. 10, the WEZ range is determined based on the blue aircraft,

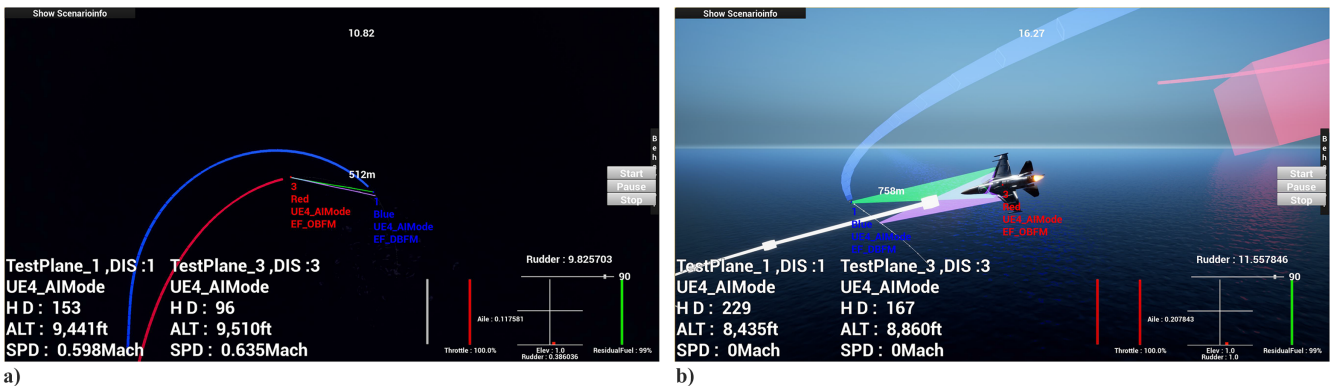


Fig. 9 Example screenshots of Unreal Engine: a) top view, and b) external view.

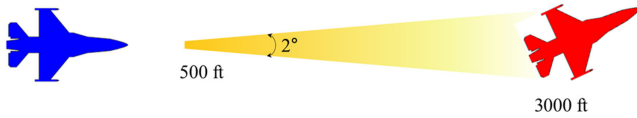


Fig. 10 Weapon engagement zone.

positioned in front of the nose, with a distance of 500–3000 ft and an angle of 2 deg. When entering the WEZ, the damage is calculated according to Eq. (2) [30]:

$$d_{wez} = \begin{cases} 0 & r > 3000 \text{ ft} \\ \frac{3000 - r}{2500} & 500 \text{ ft} \leq r \leq 3000 \text{ ft} \\ 0 & r < 500 \text{ ft} \end{cases} \quad (2)$$

where r denotes range between blue and red aircraft, and the damage decreases linearly from 500 to 3000 ft, which is set so that the damage is large at close range. Damage is calculated 60 times per second, and the damage in Eq. (2) is equivalent to 1 s of damage. For example, if the red aircraft is in the WEZ at 500 ft for 1 s, the cumulative damage is one. The engagement runs for 300 s; and if the accumulated damage during the engagement becomes one or the aircraft goes below a 1000 ft altitude, it is determined to be defeated, and the engagement is terminated [30].

B. Air-to-Air Combat Results

To evaluate the performance of the proposed method, air-to-air combat simulations are conducted. The initial setup is the HABFM, where the two aircraft start in a neutral state, as depicted in Fig. 11. The AA is set to 180 deg, the initial speed is 450 kt, and the altitude is 20,000 ft. Three different distance cases are considered between the two fighters: 2000, 2500, and 3000 ft. Additionally, the altitude and distance are randomly assigned within a range of 20 ft from the reference values. Table 1 summarizes the experimental results, where “conventional” refers to the existing algorithm models proposed in prior papers, which focus on air engagement algorithms based on the pilot’s BFM. In Refs. [3,32], they optimized the route to minimize the distance and the AA from the opposing aircraft and used the virtual pursuit point for control. “Proposed” refers to the model of the algorithm proposed in this paper. For comparison purposes, conventional is implemented in the same experimental environment, and air-to-air combat simulations are conducted. Each game result represents the average of 10 runs with the same setup. Table 1 categorizes the game types into three main categories: matches between conventional models, matches between conventional and proposed algorithms, and matches between proposed algorithms. To validate the reliability of the compared algorithm and the proposed algorithm, matches between the same models are conducted. The match results record the accumulated damage of both the red and blue aircraft until the end of the match, which are presented as percentages. The damage value of the loser in each match is displayed. RMSE stands for

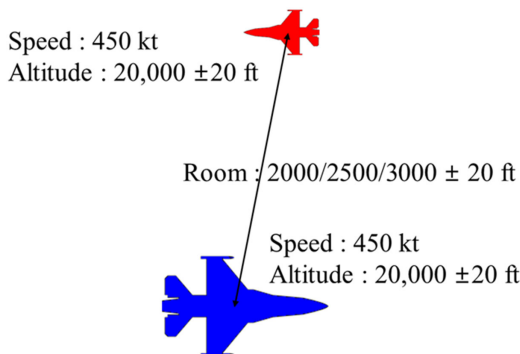


Fig. 11 Initial setup of air-to-air combat.

Table 1 Experimental results of the HABFM

Games	Room setup, ft	Damage, %	
		Red	Blue
Conventional (red aircraft) vs conventional (blue aircraft)	2000	91.8 ^a	91.5
	2500	96.0 ^a	95.2
	3000	88.1	88.8 ^a
RMSE		0.38	
Conventional (red aircraft) vs proposed (blue aircraft)	2000	100.0 ^a	88.3
	2500	90.0 ^a	84.7
	3000	100.0 ^a	65.2
RMSE		12.37	
Proposed (red aircraft) vs proposed (blue aircraft)	2000	95.1 ^a	94.4
	2500	90.6	94.1 ^a
	3000	91.7 ^a	91.6
RMSE		1.22	

^aThe damage value of the loser in each match.

root-mean-squared error and is calculated using the differences in the cumulative damage values obtained from match results in the three different settings.

The match results of the same models (conventional vs conventional or proposed vs proposed) show almost similar cumulative damage, even if the distance between the two fighters is different. Figure 12 shows the flight paths of a simulation at the setting of 2000 ft between the proposed models. In Fig. 12, the altitude is decreasing while performing two-circle maneuvers in general, and the turning radius gradually decreases. The model causes comparable damage when transitioning into a head-on engagement following the two-circle startup, and the overall outcomes show a similar pattern of accumulated damage.

However, looking at the results between the two different models, the proposed model won, regardless of the distance difference. We compare the RMSE values to show the difference more clearly. Whereas the RMSE result of conventional vs proposed is 12.37, the RMSE results between the same models are 0.38 and 1.22, respectively, with very little difference. The reason for these results can be seen by looking at Fig. 13. Figure 13 is a scene from a match where blue and red aircraft faced each other from a distance of 2500 ft. When the two fighters made a two-circle maneuver, the proposed method (blue aircraft) made a maneuver that minimized the turning radius while trying to maintain the speed, and the red aircraft made a reverse turn. It has the effect of sending the red aircraft forward of the 3–9 line. At this time, the distance between the two fighters is 7508 ft: the blue aircraft has a speed of 424 kt and an altitude of 3646 ft, and the red aircraft has a speed of 496 kt and an altitude of 2349 ft. Therefore, the difference between the two in terms of energy is not large, the distance between the two fighters is sufficient for maneuvering, and the blue aircraft is advantageous in terms of geopolitical position in the maneuvering plane; so, it receives a winning cue and becomes the OBFM mode. In summary, the blue aircraft of the proposed method optimally manages the altitude, turn speed, and turn rate according to the position of the bandit when turning; after merging, it selects the appropriate OBFM or DBFM maneuver through situational awareness, and so it wins by maintaining favorable conditions.

VI. Conclusions

This paper presents a novel autonomous maneuvering decision model derived from a basic employment manual, which outlines the necessary maneuvering procedures and techniques for completing missions in air combat. The model represents the attack–defense–neutral situations encountered in close-air combat through a continuous graph format. In unfavorable circumstances, the decision logic is applied to transition toward a dominant situation while also considering the prevailing conditions to maintain the current state and execute maneuvers to damage the enemy. To evaluate the effectiveness of the proposed model, air-to-air combat simulations are conducted against

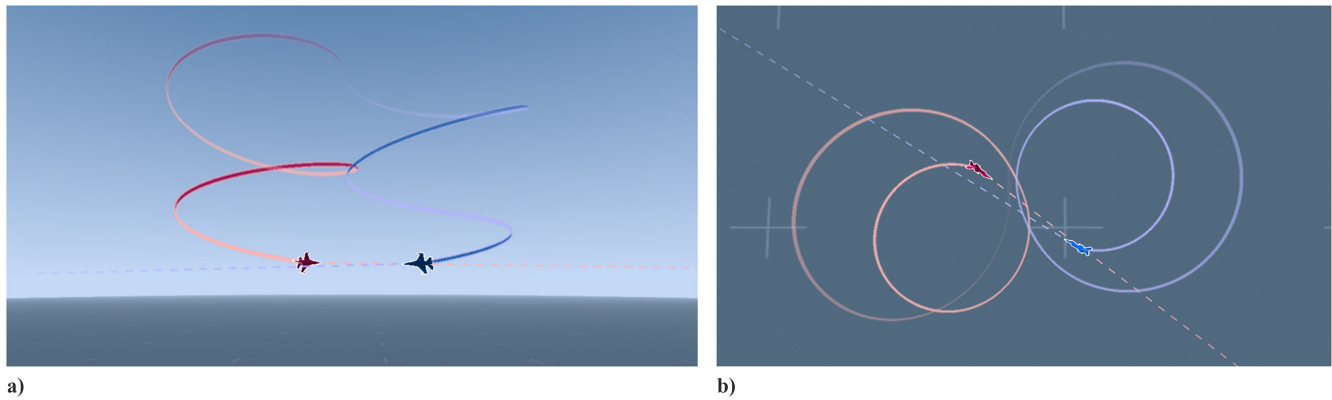


Fig. 12 Trajectories of simulation between proposed methods (2000 ft setup): a) side view, and b) top view.

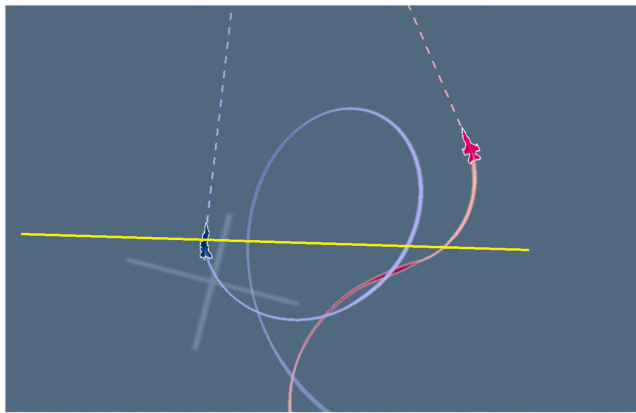


Fig. 13 Trajectories of simulation between proposed and conventional methods (2500 ft setup, where yellow straight line represents 3–9 line of the blue aircraft).

existing algorithm models under various conditions. The experimental results demonstrate that the proposed method inflicts greater damage to the adversary, regardless of the initial conditions.

Acknowledgment

This work was supported in part by the Agency for Defense Development under grant UE191120RD.

References

- [1] Isaacs, R., *Games of Pursuit*, RAND Corp., Santa Monica, CA, 1951.
- [2] *Basic Employment Manual F-16C*, Vol. 5: Korean Air Force Tactics, Techniques and Procedures 3-3, 1 Oct. 2005, <http://falcon.blu3wolf.com/Docs/Basic-Employment-Manual-F-16C-RoKAF.pdf> [retrieved 27 June 2022].
- [3] Dong, Y., Ai, J., and Liu, J., "Guidance and Control for Own Aircraft in the Autonomous Air Combat: A Historical Review and Future Prospects," *Proceedings of the Institution of Mechanical Engineers, Part G: Journal of Aerospace Engineering*, Vol. 233, No. 16, 2019, pp. 5943–5991. <https://doi.org/10.1177/0954410019889447>
- [4] Wong, R. E., "Some Aerospace Differential Games," *Journal of Spacecraft and Rockets*, Vol. 4, No. 11, 1967, pp. 1460–1465. <https://doi.org/10.2514/3.29114>
- [5] Meier, L., "A New Technique for Solving Pursuit-Evasion Differential Games," *IEEE Transactions on Automatic Control*, Vol. 14, No. 4, 1969, pp. 352–359. <https://doi.org/10.1109/TAC.1969.1099226>
- [6] Ray, W. H., and Barney, J. R., "The Application of Differential Game Theory to Process Control Problems," *Chemical Engineering Journal*, Vol. 3, Jan. 1972, pp. 237–244. [https://doi.org/10.1016/0300-9467\(72\)85027-5](https://doi.org/10.1016/0300-9467(72)85027-5)
- [7] Lynch, U. H., "Differential Game Barriers and Their Application in Air-to-Air Combat," Ph.D. Dissertation, Air Force Inst. of Technology, Wright-Patterson AFB, Dayton, OH, 1973.
- [8] Merz, A. W., "Application of Differential Game Theory to Role-Determination in Aerial Combat," NASA CR 137713, July 1975.
- [9] Järmark, B., "Near-Optimal Closed-Loop Strategy for Aerial Combat Games," Dept. of Automatic Control, Royal Inst. of Technology, 1976.
- [10] Anderson, G. M., "Feedback Control for a Pursuing Spacecraft Using Differential Dynamic Programming," *AIAA Journal*, Vol. 15, No. 8, 1977, pp. 1084–1088. <https://doi.org/10.2514/3.60758>
- [11] Horie, K., and Conway, B. A., "Optimal Fighter Pursuit-Evasion Maneuvers Found via Two-Sided Optimization," *Journal of Guidance, Control, and Dynamics*, Vol. 29, No. 1, 2006, pp. 105–112. <https://doi.org/10.2514/1.3960>
- [12] Bernhard, P., "Pursuit-Evasion Games and Zero-Sum Two-Person Differential Games," *Encyclopedia of Systems and Control*, Springer-Verlag, Berlin, 2015, pp. 1780–1785. https://doi.org/10.1007/978-1-4471-5102-9_270-1
- [13] Baron, S., and Kleinman, D. L., "A Markovian Approach to Dynamic Games of Combat," *IEEE Conference on Decision and Control*, IEEE Publ., Piscataway, NJ, 1971, pp. 464–469.
- [14] Huang, C., Dong, K., Huang, H., Tang, S., and Zhang, Z., "Autonomous Air Combat Maneuver Decision Using Bayesian Inference and Moving Horizon Optimization," *Journal of Systems Engineering and Electronics*, Vol. 29, No. 1, 2018, pp. 86–97. <https://doi.org/10.21629/JSEE.2018.01.09>
- [15] Virtanen, K., Raivio, T., and Hämäläinen, R. P., "Decision Theoretical Approach to Pilot Simulation," *Journal of Aircraft*, Vol. 36, No. 4, 1999, pp. 632–641. <https://doi.org/10.2514/2.2505>
- [16] Karelähti, J., "Modeling and On-Line Solution of Air Combat Optimization Problems and Games," Ph.D. Dissertation, Helsinki Univ. of Technology, Espoo, Finland, 2008.
- [17] McGrew, J. S., How, J. P., Williams, B., and Roy, N., "Air-Combat Strategy Using Approximate Dynamic Programming," *Journal of Guidance, Control, and Dynamics*, Vol. 33, No. 5, 2010, pp. 1641–1654. <https://doi.org/10.2514/1.46815>
- [18] Ma, Y., Ma, X., and Song, X., "A Case Study on Air Combat Decision Using Approximated Dynamic Programming," *Mathematical Problems in Engineering*, Vol. 2014, No. 4, 2014, Paper 183401. <https://doi.org/10.1155/2014/183401>
- [19] Burgin, G. H., and Owens, A. J., "An Adaptive Maneuvering Logic Computer Program for the Simulation of One-to-One Air-to-Air Combat. Volume 2: Program Description," NASA CR 2583, Sept. 1975.
- [20] Burgin, G. H., and Sidor, L. B., "Rule-Based Air Combat Simulation," No. H-1501, Sept. 1988.
- [21] Goodrich, K., and Mcmanus, J., "Development of a Tactical Guidance Research and Evaluation System (TGRES)," *Flight Simulation Technologies Conference and Exhibit*, AIAA Paper 1989-3312, 1989. <https://doi.org/10.2514/6.1989-3312>
- [22] Goodrich, K. H., "A High-Fidelity, Six-Degree-of-Freedom Batch Simulation Environment for Tactical Guidance Research and Evaluation," NASA, *Scientific and Technical Information Program*, 1993.
- [23] Lewis, M. S., and Aiken, E. W., "Piloted Simulation of One-on-One Helicopter Air Combat at NOE Flight Levels," NAS 1.15: 86686, April 1985.
- [24] Austin, F., Carbone, G., Falco, M., Hinz, H., and Lewis, M., "Automated Maneuvering Decisions for Air-to-Air Combat," *Guidance, Navigation and Control Conference*, AIAA Paper 1987-2393, 1987. <https://doi.org/10.2514/6.1987-2393>

- [25] Smith, R. E., Dike, B. A., Mehra, R. K., Ravichandran, B., and El-Fallah, A., "Classifier Systems in Combat: Two-Sided Learning of Maneuvers for Advanced Fighter Aircraft," *Computer Methods in Applied Mechanics and Engineering*, Vol. 186, Nos. 2–4, 2000, pp. 421–437.
[https://doi.org/10.1016/S0045-7825\(99\)00395-3](https://doi.org/10.1016/S0045-7825(99)00395-3)
- [26] Ernest, N., Carroll, D., Schumacher, C., Clark, M., Cohen, K., and Lee, G., "Genetic Fuzzy Based Artificial Intelligence for Unmanned Combat Aerial Vehicle Control in Simulated Air Combat Missions," *Journal of Defense Management*, Vol. 6, No. 1, 2016, pp. 2167–2173.
<https://doi.org/10.4172/2167-0374.1000144>
- [27] Yang, Q., Zhang, J., Shi, G., Hu, J., and Wu, Y., "Maneuver Decision of UAV in Short-Range Air Combat Based on Deep Reinforcement Learning," *IEEE Access*, Vol. 8, Dec. 2019, pp. 363–378.
<https://doi.org/10.1109/ACCESS.2019.2961426>
- [28] Xu, J., Guo, Q., Xiao, L., Li, Z., and Zhang, G., "Autonomous Decision-Making Method for Combat Mission of UAV Based on Deep Reinforcement Learning," *IEEE 4th Advanced Information Technology, Electronic and Automation Control Conference*, Vol. 1, IEEE, Piscataway, NJ, 2019, pp. 538–544.
<https://doi.org/10.1109/IAEAC47372.2019.8998066>
- [29] Wang, Z., Li, H., Wu, H., and Wu, Z., "Improving Maneuver Strategy in Air Combat by Alternate Freeze Games with a Deep Reinforcement Learning Algorithm," *Mathematical Problems in Engineering*, Vol. 2020, June 2020, Paper 7180639.
<https://doi.org/10.1155/2020/7180639>
- [30] Pope, A. P., Jaime, J. S., Mićović, D., Diaz, H., Rosenbluth, D., Ritholtz, L., Twedt, J. C., Waler, T. T., Alcedo, K., and Javorssek, D., "Hierarchical Reinforcement Learning for Air-to-Air Combat," *International Conference on Unmanned Aircraft Systems*, IEEE Publ., Piscataway, NJ, 2021, pp. 275–284.
<https://doi.org/10.1109/ICUAS51884.2021.9476700>
- [31] Karli, M., Efe, M. Ö., and Sever, H., "Extracting Basic Fighter Maneuvers from Actual Flight Data," *Lecture Notes on Information Theory*, Vol. 5, No. 1, 2017, pp. 1–6.
<https://doi.org/10.18178/Init.5.1.16>
- [32] You, D.-I., and Shim, H., "Design of an Aerial Combat Guidance Law Using Virtual Pursuit Point Concept," *Proceedings of the Institution of Mechanical Engineers, Part G: Journal of Aerospace Engineering*, Vol. 229, No. 5, 2015, pp. 792–813.
<https://doi.org/10.1177/0954410014539654>
- [33] Yamasaki, T., and Balakrishnan, S. N., "Sliding Mode Based Pure Pursuit Guidance for UAV Rendezvous and Chase with a Cooperative Aircraft," *IEEE American Control Conference*, IEEE Publ., Piscataway, NJ, 2010, pp. 5544–5549.
<https://doi.org/10.1109/ACC.2010.5530997>
- [34] Berndt, H., "JSBSim: An Open Source Flight Dynamics Model in C++," *AIAA Modeling and Simulation Technologies Conference and Exhibit*, AIAA Paper 2004-4923, 2004.
<https://doi.org/10.2514/6.2004-4923>
- [35] Siouris, G. M., *Missile Guidance and Control Systems*, Springer Science and Business Media, Berlin, 2004, Chap. 4.5.
<https://doi.org/10.1007/b97614>
- [36] Vogeltanz, T., "A Survey of Free Software for the Design, Analysis, Modelling, and Simulation of an Unmanned Aerial Vehicle," *Archives of Computational Methods in Engineering*, Vol. 23, No. 3, 2016, pp. 449–514.
<https://doi.org/10.1007/s11831-015-9147-y>

D. Casbeer
Associate Editor